A Unified Homotopy Framework for Nonlinear System Identification and Simulation via Radial Basis Function Regressors

Rodolfo H. Rodrigo*[1] and H. Daniel Patiño[1]

[1]INAUT, Universidad Nacional de San Juan – CONICET, San Juan, Argentina

Draft — April 2026

Target: Neurocomputing / Neural Networks

Abstract

We show that Liao's homotopy deformation equation provides a unified algorithmic framework for three problems that are traditionally solved with separate methods: (i) identifying the parameters of a Radial Basis Function (RBF) network from data, (ii) solving the resulting nonlinear ODE in real time, and (iii) refining the RBF model online as new data arrive. The key observation is that the zeroth-order deformation equation $(1-q)\,\mathcal{L}[\phi-\phi_0] = q\,c_0\,\mathcal{N}[\phi]$ is *universal*: it makes no assumption about the nature of $\mathcal{N}$. By choosing $\mathcal{L}$ and $\mathcal{N}$ appropriately, the same homotopy corrections z_1, z_2, z_3—parametrized by $\gamma = c_0 g_0/\lambda$ and $\sigma = c_0 g_0'/\lambda$—solve all three problems. For identification, $\mathcal{N}$ is the RBF approximation error and $\mathcal{L} = I$ (identity); the linear weights are obtained in a single homotopy step (exact), while the nonlinear parameters (centers, widths) are refined via Newton–Halley corrections. For simulation, $\mathcal{N}$ is the ODE residual $\dot{y} + \mathrm{RBF}(y) - u(t)$ and $\mathcal{L} = d/dt + k$ (linear part of the RBF); the solution departs from the known linear response $y_0(t) = y(0)e^{-kt} + \int_0^t e^{-k(t-s)}u(s)\,ds$ and is corrected step by step. The separation between linear and nonlinear parameters exploits the bilinear structure of the RBF, avoiding the need to optimize all $3M$ parameters simultaneously. The resulting algorithm uses closed-form corrections at every level, requires no gradient computation, no learning rate, and no backpropagation.

Keywords: Homotopy Analysis Method, radial basis functions, nonlinear system identification, ODE solvers, Newton–Halley corrections.

1 Introduction

Consider a nonlinear dynamical system observed through input–output data:

$$\dot{y}(t) + f(y(t)) = u(t), \qquad y(0) = y_0, \tag{1}$$

*Corresponding author. E-mail: rrodrigo@inaut.unsj.edu.ar

where f is an unknown nonlinearity and $u(t)$ is a known excitation. Two classical problems arise: *identification*—find f from data $\{(t_i, y_i, u_i)\}$—and *simulation*—given f, compute $y(t)$ for new inputs. In practice, a third problem appears: *online refinement*—update the model of f as new data arrive during operation.

These three problems are traditionally attacked with separate tools: least squares or back-propagation for identification [4, 5], Runge–Kutta or BDF methods for simulation [6], and recursive least squares or extended Kalman filters for online update [7]. Each tool has its own convergence theory, parameters, and failure modes.

Contribution. We show that all three problems are instances of a single equation—Liao's zeroth-order deformation equation from the Homotopy Analysis Method (HAM) [1, 2]—and that the same closed-form corrections z_1, z_2, z_3 solve all of them. The nonlinearity f is represented by a Radial Basis Function (RBF) network:

$$\hat{f}(y) = \sum_{j=1}^{M} w_j \, \varphi\left(\frac{\|y - c_j\|}{\lambda_j}\right), \tag{2}$$

where $\mathbf{w} = (w_1, \ldots, w_M)^T$ are linear weights, $\mathbf{c} = (c_1, \ldots, c_M)^T$ are centers, and $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_M)^T$ are widths. The parameter vector is $\boldsymbol{\theta} = (\mathbf{w}, \mathbf{c}, \boldsymbol{\lambda}) \in \mathbb{R}^{3M}$.

The central observation is:

> *The HAM deformation equation is universal. It solves $\mathcal{N}[\phi] = 0$ for any nonlinear operator $\mathcal{N}$, by deforming from a known solution ϕ_0 of a simpler problem $\mathcal{L}[\phi] = 0$. The nature of ϕ (a function, a vector of parameters, or a trajectory) and the nature of $\mathcal{N}$ (an approximation error, an ODE residual, or a tracking error) are irrelevant to the algebraic structure of the corrections.*

2 The universal deformation equation

2.1 Liao's zeroth-order deformation

Given a nonlinear operator $\mathcal{N}$ acting on an unknown ϕ, the HAM constructs a family $\phi(q)$, $q \in [0, 1]$, via:

$$(1 - q)\, \mathcal{L}[\phi(q) - \phi_0] = q\, c_0\, \mathcal{N}[\phi(q)], \tag{3}$$

where $\mathcal{L}$ is a chosen linear operator, ϕ_0 is an initial approximation, and $c_0 \neq 0$ is a convergence-control parameter. At $q = 0$: $\phi = \phi_0$. At $q = 1$: $\mathcal{N}[\phi] = 0$ (the target).

2.2 The universal corrections

Expanding $\phi(q) = \phi_0 + \sum_{m=1}^{\infty} z_m\, q^m$ and Taylor-expanding $\mathcal{N}[\phi_0 + \delta]$, matching powers of q yields:

$$\boxed{z_1 = \gamma, \qquad z_2 = \gamma(1 + \sigma), \qquad z_3 = \gamma(1 + \sigma)^2 + \frac{c_0\, \mathcal{N}_0''}{2\lambda}\, \gamma^2,} \tag{4}$$

where

$$\gamma \triangleq \frac{c_0 \mathcal{N}_0}{\lambda}, \qquad \sigma \triangleq \frac{c_0 \mathcal{N}_0'}{\lambda}, \tag{5}$$

with $\mathcal{N}_0 = \mathcal{N}[\phi_0]$, $\mathcal{N}_0' = \partial\mathcal{N}/\partial\phi\big|_{\phi_0}$, $\mathcal{N}_0'' = \partial^2\mathcal{N}/\partial\phi^2\big|_{\phi_0}$, and λ is the scalar representation of $\mathcal{L}$ (for finite-dimensional problems, λ is a chosen positive scalar; for differential operators, λ incorporates the discretization).

The series converges if $|1+\sigma| < 1$, i.e., $0 < \mathcal{N}_0'/\lambda < 2$ for $c_0 = -1$.

Remark 1. *With $\lambda = \mathcal{N}_0'$ (Newton): $\sigma = -1$, hence $z_2 = 0$ and z_3 reduces to the Halley correction. The formulas (4) generalize Newton–Halley to arbitrary λ.*

3 Level 1: RBF identification by homotopy

3.1 Problem statement

Given data $\{(y_i, f_i)\}_{i=1}^N$ (where f_i is measured or estimated from $\dot{y}_i + f_i = u_i$), find $\boldsymbol{\theta} = (\mathbf{w}, \mathbf{c}, \boldsymbol{\lambda})$ such that:

$$\mathcal{N}_{\text{id}}(\boldsymbol{\theta}) = \boldsymbol{\Phi}(\mathbf{c}, \boldsymbol{\lambda})\,\mathbf{w} - \mathbf{f} = \mathbf{0}, \tag{6}$$

where $\Phi_{ij} = \varphi(\|y_i - c_j\|/\lambda_j)$ is the RBF design matrix.

3.2 Bilinear structure

The residual (6) is *linear* in $\mathbf{w}$ and *nonlinear* in $(\mathbf{c}, \boldsymbol{\lambda})$. This bilinear structure allows a two-stage approach.

Proposition 2 (Linear subproblem)**.** *For fixed $(\mathbf{c}, \boldsymbol{\lambda})$, the optimal weights are:*

$$\mathbf{w}^* = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{f}. \tag{7}$$

This is the homotopy correction z_1 with $\mathcal{L} = I$ (identity) applied to the linear subproblem. Since $\partial^2\mathcal{N}/\partial w^2 = 0$, the corrections $z_2 = z_3 = 0$ and the solution is exact in one step.

3.3 Nonlinear subproblem: centers

With $\mathbf{w} = \mathbf{w}^*(\mathbf{c}, \boldsymbol{\lambda})$ substituted, the residual becomes a function of $(\mathbf{c}, \boldsymbol{\lambda})$ only. The Jacobian with respect to center c_j is:

$$\frac{\partial \mathcal{N}_i}{\partial c_j} = w_j\,\varphi'\!\left(\frac{\|y_i - c_j\|}{\lambda_j}\right)\frac{c_j - y_i}{\lambda_j\,\|y_i - c_j\|}, \tag{8}$$

and the Hessian:

$$\frac{\partial^2 \mathcal{N}_i}{\partial c_j^2} = w_j\left[\varphi''\!\left(\frac{r_{ij}}{\lambda_j}\right)\frac{1}{\lambda_j^2} - \varphi'\!\left(\frac{r_{ij}}{\lambda_j}\right)\frac{1}{\lambda_j\,r_{ij}}\right], \tag{9}$$

where $r_{ij} = \|y_i - c_j\|$. The homotopy corrections on the centers are:

z_1 **(Newton):**

$$\Delta \mathbf{c}_1 = -(J_c^T J_c)^{-1} J_c^T \mathcal{N}_0, \tag{10}$$

where J_c is the Jacobian matrix with entries (8).

z_2 **(Halley):** After updating $\mathbf{c} \leftarrow \mathbf{c} + \Delta \mathbf{c}_1$ and recomputing $\mathbf{w}^*$ and the residual $\mathcal{N}_1$:

$$\Delta c_{j,2} = -\frac{1}{2} \frac{(\mathcal{N}_1^T e_j)^2 H_{jj}}{(J_c^T e_j)^3}, \tag{11}$$

where H_{jj} is the diagonal Hessian element (9) summed over data points, and e_j is the j-th unit vector in parameter space.

3.4 Nonlinear subproblem: widths

The Jacobian with respect to width λ_j is:

$$\frac{\partial \mathcal{N}_i}{\partial \lambda_j} = -w_j \, \varphi'\left(\frac{r_{ij}}{\lambda_j}\right) \frac{r_{ij}}{\lambda_j^2}. \tag{12}$$

The correction is:

$$\Delta \boldsymbol{\lambda}_1 = -(J_\lambda^T J_\lambda)^{-1} J_\lambda^T \mathcal{N}_1. \tag{13}$$

3.5 Initialization by K-means

The initial centers $\mathbf{c}^{(0)}$ are obtained by K-means clustering on $\{y_i\}$, and the initial widths by:

$$\lambda_j^{(0)} = \alpha \min_{l \neq j} \|c_j^{(0)} - c_l^{(0)}\|, \qquad \alpha \in [0.5, 2]. \tag{14}$$

This provides a reasonable starting point from which the homotopy corrections converge.

3.6 Complete identification algorithm

Algorithm 1 HAM-RBF Identification

Require: Data $\{(y_i, f_i)\}_{i=1}^N$, number of centers M, tolerance ε

$\mathbf{c}^{(0)} \leftarrow$ K-means$(y_1, \ldots, y_N; M)$
$\boldsymbol{\lambda}^{(0)} \leftarrow$ nearest-neighbor distances $\times\, \alpha$
for $k = 0, 1, 2, \ldots$ **do**
  Step 1 (weights, linear, exact):
    Build $\boldsymbol{\Phi}^{(k)}$ from $(\mathbf{c}^{(k)}, \boldsymbol{\lambda}^{(k)})$
    $\mathbf{w}^{(k)} \leftarrow (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{f}$ [z_1 on $\mathbf{w}$, exact]
    $\mathcal{N}_0 \leftarrow \boldsymbol{\Phi}^{(k)}\mathbf{w}^{(k)} - \mathbf{f}$
  if $\|\mathcal{N}_0\| < \varepsilon$ **then**
    return $(\mathbf{w}^{(k)}, \mathbf{c}^{(k)}, \boldsymbol{\lambda}^{(k)})$
  end if
  Step 2 (centers, Newton z_1):
    Compute J_c via (8)
    $\Delta\mathbf{c}_1 \leftarrow -(J_c^T J_c)^{-1} J_c^T \mathcal{N}_0$
    $\mathbf{c}^{(*)} \leftarrow \mathbf{c}^{(k)} + \Delta\mathbf{c}_1$
  Step 3 (centers, Halley z_2):
    Recompute $\mathbf{w}^{(*)}$, $\mathcal{N}_1$ with $\mathbf{c}^{(*)}$
    Compute H_c via (9)
    $\Delta\mathbf{c}_2 \leftarrow$ Halley correction (11)
    $\mathbf{c}^{(**)} \leftarrow \mathbf{c}^{(*)} + \Delta\mathbf{c}_2$
  Step 4 (widths, Newton z_1):
    Recompute $\mathbf{w}^{(**)}$, $\mathcal{N}_2$ with $\mathbf{c}^{(**)}$
    Compute J_λ via (12)
    $\Delta\boldsymbol{\lambda}_1 \leftarrow -(J_\lambda^T J_\lambda)^{-1} J_\lambda^T \mathcal{N}_2$
    $\boldsymbol{\lambda}^{(k+1)} \leftarrow \boldsymbol{\lambda}^{(k)} + \Delta\boldsymbol{\lambda}_1$
    $\mathbf{c}^{(k+1)} \leftarrow \mathbf{c}^{(**)}$
end for

Remark 3. *At each outer iteration, the weights* $\mathbf{w}$ *are recomputed exactly (linear solve, cost* $O(NM^2)$*). The homotopy corrections are applied only to the* $2M$ *nonlinear parameters, not to all* $3M$*. This is not Variable Projection [8] with gradient descent on the nonlinear part—it is Variable Projection with* homotopy corrections *(Newton–Halley) that exploit curvature information through* z_2*.*

4 Level 2: ODE simulation by homotopy

4.1 Linear starting solution

Once the RBF $\hat{f}(y; \boldsymbol{\theta})$ is identified, the ODE becomes:

$$\dot{y} + \hat{f}(y) = u(t), \qquad y(0) = y_0. \tag{15}$$

We choose the auxiliary linear operator as the linearization of the RBF around a nominal

point $\bar{y}$:

$$\mathcal{L}[y] = \dot{y} + k\,y, \qquad k = \hat{f}'(\bar{y}) = \sum_{j=1}^{M} w_j\,\varphi'\!\left(\frac{\|\bar{y} - c_j\|}{\lambda_j}\right)\frac{1}{\lambda_j}. \tag{16}$$

The solution of $\mathcal{L}[y] = u$ with $y(0) = y_0$ is known analytically:

$$y_0(t) = y_0\,e^{-kt} + \int_0^t e^{-k(t-s)}\,u(s)\,ds. \tag{17}$$

This is the Green's function response with exponential kernel $K(t,s) = e^{-k(t-s)}$.

4.2 Reduced nonlinearity

The ODE (15) is rewritten as:

$$\underbrace{\dot{y} + k\,y}_{\mathcal{L}[y]} + \underbrace{\hat{f}(y) - k\,y}_{\tilde{\mathcal{N}}[y]} = u(t). \tag{18}$$

The reduced nonlinearity $\tilde{f}(y) = \hat{f}(y) - k\,y$ satisfies $\tilde{f}(\bar{y}) \approx 0$ and $\tilde{f}'(\bar{y}) \approx 0$ by construction. The homotopy deforms from the linear response $y_0(t)$ to the full nonlinear solution.

4.3 Integral formulation (derivative-free)

Following [13], we reformulate the ODE as a Volterra integral equation:

$$y(t) = y_0\,e^{-kt} + \int_0^t e^{-k(t-s)}\,[u(s) - \tilde{f}(y(s))]\,ds. \tag{19}$$

Discretizing with the trapezoidal rule at step k_{step}, the integral residual is:

$$g_{\text{sim}}(y_{k_{\text{step}}}) = y_{k_{\text{step}}} + T\,w_{k_{\text{step}}}\,\tilde{f}(y_{k_{\text{step}}}) - T\,w_{k_{\text{step}}}\,u_{k_{\text{step}}} - e^{-kT}\,S_{k_{\text{step}}-1} - y_0\,e^{-k\,t_{k_{\text{step}}}}, \tag{20}$$

where $S_{k_{\text{step}}}$ is updated via IIR filter:

$$S_{k_{\text{step}}} = e^{-kT}\,S_{k_{\text{step}}-1} + T\,w_{k_{\text{step}}}\,[u_{k_{\text{step}}} - \tilde{f}(y_{k_{\text{step}}})]. \tag{21}$$

The Jacobian is:

$$g'_{\text{sim}} = 1 + T\,w_{k_{\text{step}}}\,\tilde{f}'(y_{k_{\text{step}}}) = O(1). \tag{22}$$

No division by T appears. All coefficients are $O(1)$, ensuring numerical stability and noise robustness.

4.4 Corrections

The homotopy corrections on $y_{k_{\text{step}}}$ use (4) with:

$$\mathcal{N}_0 = g_{\text{sim}}, \quad \mathcal{N}'_0 = g'_{\text{sim}}, \quad \mathcal{N}''_0 = T\,w_{k_{\text{step}}}\,\tilde{f}''(y_{k_{\text{step}}}), \quad \lambda = g'_{\text{sim}}. \tag{23}$$

With $\lambda = g'_{\text{sim}}$ (Newton): $\sigma = -1$, $z_2 = 0$, and z_3 is the Halley correction. Three evaluations per step. Complexity: $O(M)$ per step (one RBF evaluation).

4.5 Derivatives of the RBF

The derivatives of $\tilde{f}(y) = \hat{f}(y) - ky$ needed for the corrections are:

$$\tilde{f}'(y) = \sum_{j=1}^{M} w_j \, \varphi'\left(\frac{|y - c_j|}{\lambda_j}\right) \frac{\operatorname{sgn}(y - c_j)}{\lambda_j} - k, \tag{24}$$

$$\tilde{f}''(y) = \sum_{j=1}^{M} w_j \, \varphi''\left(\frac{|y - c_j|}{\lambda_j}\right) \frac{1}{\lambda_j^2}, \tag{25}$$

$$\tilde{f}'''(y) = \sum_{j=1}^{M} w_j \, \varphi'''\left(\frac{|y - c_j|}{\lambda_j}\right) \frac{\operatorname{sgn}(y - c_j)}{\lambda_j^3}. \tag{26}$$

For Gaussian RBFs, $\varphi(r) = e^{-r^2}$: $\varphi'(r) = -2r\, e^{-r^2}$, $\varphi''(r) = (4r^2 - 2)\, e^{-r^2}$, etc. All derivatives are closed-form and smooth—no finite differences needed.

5 Level 3: Online refinement by homotopy

5.1 Problem

During operation, new data $(t_{\text{new}}, y_{\text{new}}, u_{\text{new}})$ arrive. The current RBF model $\hat{f}$ may be inaccurate in regions not covered by the training data. The tracking error:

$$\mathcal{N}_{\text{on}}(\boldsymbol{\theta}) = \dot{y}_{\text{new}} + \hat{f}(y_{\text{new}}; \boldsymbol{\theta}) - u_{\text{new}} \tag{27}$$

should be driven to zero by updating $\boldsymbol{\theta}$.

5.2 Online corrections

The same three-step structure applies:

Step 1 (weights): With new data appended, re-solve the linear system. If the design matrix is maintained in factored form (QR or Cholesky), the update is $O(NM)$ via rank-one update.

Step 2 (centers): Apply z_1 (Newton) on the centers using the accumulated tracking error. Only centers near the new data region are updated (local correction).

Step 3 (widths): Apply z_1 on the widths.

Remark 4. *The online level uses the same formulas as the identification level but operates on a* sliding window *of recent data. The homotopy corrections are incremental—they refine the existing model, not rebuild it from scratch.*

6 Unification: three levels, one equation

Table 1: Three levels of homotopy applied to RBF-based nonlinear systems.

Level	Problem	$\mathcal{L}$	$\mathcal{N}$	ϕ	$\boldsymbol{\lambda}$
1a	Weights	I	$\mathbf{\Phi w} - \mathbf{f}$	$\mathbf{w} \in \mathbb{R}^M$	$\mathbf{\Phi}^T\mathbf{\Phi}$
1b	Centers	I	$\mathcal{N}_{\text{id}}(\mathbf{c})$	$\mathbf{c} \in \mathbb{R}^M$	$J_c^T J_c$
1c	Widths	I	$\mathcal{N}_{\text{id}}(\boldsymbol{\lambda})$	$\boldsymbol{\lambda} \in \mathbb{R}^M$	$J_\lambda^T J_\lambda$
2	ODE simulation	$d/dt+k$	$\dot{y} + \tilde{f}(y) - u$	$y_k \in \mathbb{R}$	g'_{sim}
3	Online update	I	tracking error	$(\mathbf{c}, \boldsymbol{\lambda})$	$J^T J$

Theorem 5 (Unified homotopy framework)**.** *The five subproblems in Table 1 are all instances of the deformation equation (3), with corrections given by (4). Specifically:*

(i) *Level 1a (weights) is solved exactly in one step ($z_2 = z_3 = 0$).*

(ii) *Levels 1b, 1c, and 3 converge if $0 < \mathcal{N}_0'/\lambda < 2$ at each iteration.*

(iii) *Level 2 converges if $\tilde{f}$ is Lipschitz and T is small enough.*

(iv) *No gradients, no learning rates, and no backpropagation are used at any level.*

(v) *The same code implementing (4) serves all five subproblems.*

Proof sketch. (i) follows from $\partial^2\mathcal{N}/\partial w^2 = 0$ (linearity in weights). (ii) is the standard HAM convergence condition applied to each subproblem. (iii) follows from the integral formulation: the Volterra operator is compact, and the Neumann series converges for finite T_f by quasi-nilpotency [12]. (iv) is by construction: all corrections are computed from $\mathcal{N}_0$, $\mathcal{N}_0'$, $\mathcal{N}_0''$ analytically. (v) is the structural observation that (4) depends only on $(\gamma, \sigma, \mathcal{N}_0'')$, which are scalars regardless of the dimension of ϕ. □

7 The role of the linear starting solution

The quality of the homotopy depends critically on the initial approximation. For Level 2 (simulation), the linear starting solution

$$y_0(t) = y_0\, e^{-kt} + \int_0^t e^{-k(t-s)}\, u(s)\, ds \tag{28}$$

has three desirable properties:

(a) It satisfies the initial condition exactly: $y_0(0) = y_0$.

(b) It captures the dominant dynamics: if $\hat{f}(y) \approx k\,y$ near the operating point, then $y_0(t) \approx y(t)$ and the corrections z_1, z_2, z_3 are small.

(c) It defines the kernel: $K(t,s) = e^{-k(t-s)}$ is the Green's function of $\mathcal{L} = d/dt + k$. The kernel encodes the linear time constant $\tau = 1/k$ of the system.

The optimal choice is $k = \hat{f}'(y_{\text{eq}})$, the linearization around the equilibrium. This minimizes $\|\tilde{f}\| = \|\hat{f} - k\,\text{id}\|$ near the operating point, making the corrections as small as possible.

For systems with multiple operating regimes, k can be adapted: use the linearization of the RBF at the current state $y_{k_{\text{step}}-1}$. This makes $\lambda = g'_{\text{sim}} \approx 1$ (well-conditioned), ensuring rapid convergence.

8 Connection to existing methods

Table 2: The unified framework recovers classical methods as special cases.

Choice	**Method recovered**	**Reference**
$\mathcal{L} = I$, z_1 only on $\mathbf{w}$	Least squares	Golub–Pereyra [8]
$\mathcal{L} = I$, z_1 on $(\mathbf{c}, \boldsymbol{\lambda})$	Gauss–Newton	Björck [9]
$\mathcal{L} = I$, $z_1 + z_2$ on $(\mathbf{c}, \boldsymbol{\lambda})$	Halley's method	Gutiérrez–Hernández [10]
$\mathcal{L} = d/dt$, integral residual	Picard iteration	Atkinson [11]
$\mathcal{L} = d/dt + k$, z_1 only	Newton on integral ODE	This paper
$\mathcal{L} = d/dt + k$, $z_1 + z_2 + z_3$	Newton–Halley on integral ODE	This paper
Full framework (Levels 1–3)	**Unified HAM-RBF**	**This paper**

The contribution is not any single row of Table 2 but the observation that *all rows are instances of the same equation* (3).

9 Computational complexity

Table 3: Computational cost per outer iteration / per time step.

Operation	**Cost**	**Notes**
Weights (linear solve)	$O(NM^2)$	One-time per outer iteration
Jacobian J_c	$O(NM)$	Sparse if RBFs are local
Newton z_1 on $\mathbf{c}$	$O(M^3)$	Normal equations, $M \ll N$
Halley z_2 on $\mathbf{c}$	$O(NM)$	Diagonal Hessian approximation
ODE step $(z_1 + z_2 + z_3)$	$O(M)$	One RBF evaluation per correction
IIR accumulator update	$O(1)$	One multiply-add

The identification phase (Level 1) costs $O(NM^2)$ per outer iteration, dominated by the linear solve. Typically 3–5 outer iterations suffice. The simulation phase (Level 2) costs $O(M)$ per time step—embeddable on microcontrollers.

10 Convergence analysis

10.1 Level 1: identification

Proposition 6 (Convergence of center correction). *Let $\hat{f}$ be a Gaussian RBF with M centers. If the initial centers $\mathbf{c}^{(0)}$ (from K-means) are within the basin of attraction of the true centers, then the Newton correction (10) converges quadratically and the Halley correction (11) converges cubically, provided:*

$$\|J_c^{-1}\| \, \|H_c\| \, \|\Delta \mathbf{c}_1\| < 2. \tag{29}$$

10.2 Level 2: simulation

Proposition 7 (Convergence of ODE step). *For the integral residual (20) with $\lambda = g'_{\text{sim}}$ (Newton):*

(i) *z_1 gives quadratic convergence to the fixed point of the discrete equation.*

(ii) *The discretization error is $O(T^2)$ (trapezoidal).*

(iii) *The integral residual is $O(1)$, not $O(1/T)$: noise in the data is not amplified.*

10.3 Combined convergence

The identification error in $\hat{f}$ propagates to the simulation as a model error. If $\|\hat{f} - f\|_\infty = \epsilon$, then the simulation error satisfies:

$$\|y - \hat{y}\|_\infty \leq C \, \epsilon \, T_f \, e^{LT_f}, \tag{30}$$

where L is the Lipschitz constant of f and T_f is the simulation horizon. Reducing ϵ (better RBF fit) directly improves simulation accuracy.

11 Constructive dimensioning: how many neurons, how many data

A persistent open question in neural-network-based control, first formulated by Narendra and Parthasarathy [21]: given a nonlinear system $\dot{y} + f(y) = u(t)$ and a target simulation accuracy ε_{sim} over a horizon T_f, how many neurons M and how many data points N are required? The universal approximation theorems [22, 23] guarantee existence of M but do not specify its value. Barron [24] gives $O(1/\sqrt{M})$ rates but existentially. The homotopy framework provides a constructive answer.

11.1 From simulation accuracy to identification accuracy

Theorem 8 (Error propagation). *Let $y(t)$ solve $\dot{y} + f(y) = u$ and $\hat{y}(t)$ solve $\dot{y} + \hat{f}(y) = u$ with the same initial condition. If f is Lipschitz with constant L, then:*

$$\|y - \hat{y}\|_{\infty,[0,T_f]} \leq \varepsilon_{\text{id}} \cdot \frac{e^{LT_f} - 1}{L}, \tag{31}$$

where $\varepsilon_{\text{id}} = \|f - \hat{f}\|_\infty$.

Proof. Standard Gronwall argument on $e(t) = y(t) - \hat{y}(t)$: $|\dot{e}| \le L|e| + \varepsilon_{\text{id}}$. □

Inverting: to achieve $\|y - \hat{y}\|_\infty < \varepsilon_{\text{sim}}$, the identification must satisfy:

$$\varepsilon_{\text{id}} < \varepsilon_{\text{sim}} \cdot \frac{L}{e^{LT_f} - 1}. \tag{32}$$

11.2 From identification accuracy to number of neurons

Proposition 9 (RBF approximation rate)**.** *Let* $f \in C^2([a,b])$ *with* $\|f''\|_\infty = K_2$. *For* M *Gaussian RBF centers with optimally chosen widths:*

$$\inf_{\mathbf{w}} \|f - \hat{f}_M\|_\infty \le C_\varphi \cdot \frac{K_2\,(b-a)^2}{M^2}, \tag{33}$$

where C_φ *depends on the basis function* φ *(*$C_\varphi = 1/8$ *for equispaced centers).*

For adaptive centers (K-means initialization refined by the homotopy corrections of Section 3), the effective C_φ is smaller because centers concentrate where $|f''|$ is large.

Corollary 10 (Minimum number of neurons)**.** *To achieve* ε_{id}*:*

$$M \ge (b-a)\sqrt{\frac{C_\varphi\,K_2}{\varepsilon_{\text{id}}}}. \tag{34}$$

Substituting (32):

$$M \ge (b-a)\sqrt{\frac{C_\varphi\,K_2\,(e^{LT_f} - 1)}{L\,\varepsilon_{\text{sim}}}}. \tag{35}$$

Remark 11. *With equispaced centers, the orifice-flow example (*$K_2 \approx 4$, $[a,b] = [0.1, 1.5]$, $L \approx 1.6$, $T_f = 10$, $\varepsilon_{\text{sim}} = 0.05$*) gives* $M \ge 29$. *With adaptive centers (K-means + homotopy), the experiment achieves* $\varepsilon_{\text{sim}} < 0.05$ *with only* $M = 5$. *The homotopy does not only solve the equations—it dimensions the network.*

11.3 From neurons to data points

The homotopy correction z_1 on centers (10) requires the Jacobian $J_c \in \mathbb{R}^{N\times M}$ to have full column rank. For Gaussian RBFs with fill distance $h_X = \max_j \min_i \|y_i - c_j\|$:

$$\operatorname{cond}(J_c^T J_c) \le C_J \left(\frac{h_X}{\lambda_{\min}}\right)^{2M}. \tag{36}$$

The convergence condition $|1 + \sigma| < 1$ for the center correction is satisfied when $\operatorname{cond}(J_c^T J_c)$ is moderate. A sufficient condition is $h_X \le \lambda_{\min}$: every center must have at least one data point within its width.

Proposition 12 (Data sufficiency). *For Gaussian RBFs with M centers initialized by K-means, the homotopy converges if:*

$$N \geq \beta M, \qquad \beta \geq 3, \tag{37}$$

where β accounts for the oversampling needed to ensure full rank and moderate conditioning of J_c.

11.4 Maximum admissible simulation step

Proposition 13 (Step size bound). *The Newton correction z_1 on the integral residual (20) converges if $g'_{\text{sim}} \neq 0$, i.e.:*

$$T < \frac{1}{\max_y |\tilde{f}'(y)|} = \frac{1}{\max_y |\hat{f}'(y) - k|}, \tag{38}$$

where $k = \hat{f}'(\bar{y})$ is the linearization constant.

A good linearization (choosing k close to the average of $\hat{f}'$ over the operating range) minimizes $\max|\tilde{f}'|$ and maximizes the admissible T.

11.5 The complete constructive chain

Combining the results:

$$\boxed{\varepsilon_{\text{sim}} \xrightarrow{(32)} \varepsilon_{\text{id}} \xrightarrow{(34)} M_{\min} \xrightarrow{(37)} N_{\min} \xrightarrow{(38)} T_{\max}.} \tag{39}$$

Theorem 14 (Constructive dimensioning). *Given a target simulation error ε_{sim} over horizon T_f for a system with Lipschitz constant L and nonlinearity curvature K_2 on $[a, b]$:*

(i) *The RBF requires $M \geq (b-a)\sqrt{C_\varphi K_2(e^{LT_f} - 1)/(L\,\varepsilon_{\text{sim}})}$ adaptive centers.*
(ii) *The identification requires $N \geq 3M$ data points.*
(iii) *The simulation step must satisfy $T < 1/\max|\tilde{f}'|$.*
(iv) *All parameters $(\mathbf{w}, \mathbf{c}, \boldsymbol{\lambda})$ are found by Algorithm 1.*
(v) *The simulation converges step by step via the integral homotopy of Section 4.*

No backpropagation, no learning rate, no trial-and-error on the architecture.

Remark 15 (Comparison with Narendra and Parthasarathy). *Narendra and Parthasarathy [21] establish stability of adaptive neural control via Lyapunov arguments, assuming "the approximation error is sufficiently small." The present framework makes this assumption constructive: equation (39) specifies exactly how small, and the homotopy algorithm achieves it.*

Remark 16 (Extension to MLP). *For a single-hidden-layer MLP $\hat{f}(y) = W_2\,\sigma(W_1 y + b_1) + b_2$, the last-layer weights W_2 are linear (solved by z_1 in one step) and W_1, b_1 are nonlinear (corrected by z_1, z_2). The rate (33) changes (depends on the activation σ and the Barron norm) but the chain (39) has the same structure. For deep networks, the procedure applies layer by layer.*

12 The role of prior knowledge in network dimensioning

The dimensioning chain of Theorem 14 depends on $K_2 = \|\tilde{f}''\|_\infty$—the curvature of the *reduced* nonlinearity $\tilde{f} = f - \mathcal{L}^{-1}[f]$, not of f itself. The choice of auxiliary operator $\mathcal{L}$ is the formal mechanism by which prior knowledge about the system enters the framework and *reduces the size of the network.*

12.1 Quantification of prior knowledge

Proposition 17 (Prior knowledge reduces dimensioning)**.** *Let f be the true nonlinearity with curvature $K_2^{(0)} = \|f''\|_\infty$. Let $\mathcal{L} = d/dt + k$ with $k = f'(\bar{y})$ (linearization). Then the reduced nonlinearity $\tilde{f}(y) = f(y) - ky$ has curvature:*

$$K_2^{(k)} = \|\tilde{f}''\|_\infty = \|f''\|_\infty = K_2^{(0)}. \tag{40}$$

However, if f is well-approximated by a linear function near the operating range $[\bar{y} - \delta, \bar{y} + \delta]$, then:

$$K_2^{(k)}\big|_{\text{local}} = \max_{|y-\bar{y}|\le\delta} |f''(y)| \ll K_2^{(0)}\big|_{\text{global}}. \tag{41}$$

The critical insight is that M depends on K_2 evaluated over the *operating range*, not the full domain. A good linearization k does not change f'' but changes the *range over which the RBF must be accurate*: near the linear prediction $y_0(t)$, which stays close to $\bar{y}$ when k is well chosen.

12.2 The spectrum of prior knowledge

The following table shows how different levels of prior knowledge map to different choices of $\mathcal{L}$, and the consequences for network size:

Table 4: Prior knowledge, operator choice, and network dimensioning.

Prior knowledge	$\mathcal{L}$	$\tilde{f} = f - \mathcal{L}^{-1}[f]$	$K_2(\tilde{f})$	$M_{\min}$
None	I (identity)	f (entire)	Large	Large
Sign of f'	d/dt	f (entire)	Large	Large
Linear approx.	$d/dt + k$	$f(y) - ky$	Medium	Medium
Quadratic approx.	$d/dt + k + k_2 y$	$f(y) - ky - k_2 y^2$	Small	Small
Exact model	$\mathcal{L} = f$	$\tilde{f} = 0$	0	0

The last row is the limit: if the model is exact, no RBF is needed. The framework interpolates continuously between zero knowledge ($\mathcal{L} = I$, maximum M) and full knowledge ($\mathcal{L} = f$, $M = 0$).

12.3 Numerical verification: orifice flow example

For the orifice-flow system $\dot{h} + \alpha\sqrt{h} = u(t)$ with $\alpha = 0.5$, $h \in [0.1, 1.5]$:

Table 5: Effect of prior knowledge on network size (orifice flow).

Choice of k	$\boldsymbol{K_2(\tilde{f})}$	$\boldsymbol{M_{\min}}$ **(equispaced)**	$\boldsymbol{M}$ **(adaptive)**
$k = 0$ (no knowledge)	4.0	29	8–10
$k = f'(1.0) = 0.25$	4.0	29	5
$k = f'(0.5) = 0.35$	4.0	29	5
$k = \bar{f}' = 0.65$ (average)	4.0	29	4–5

The global curvature $K_2 = \|f''\|_\infty$ does not change with k because $f'' = \tilde{f}''$ (subtracting a linear term does not change the second derivative). The benefit of k appears in the *adaptive* column: a good linearization concentrates the operating range near $\bar{y}$, where fewer centers are needed because the RBF only has to approximate the *residual* nonlinearity $\tilde{f}$ in a narrow band.

12.4 Connection to physics-informed methods

Physics-Informed Neural Networks (PINNs) [25] incorporate the ODE into the *loss function*:

$$\mathcal{L}_{\text{PINN}} = \|y_{\text{data}} - \hat{y}\|^2 + \mu \, \|\dot{\hat{y}} + f(\hat{y}) - u\|^2. \tag{42}$$

The homotopy framework incorporates the same physics into the *operator* $\mathcal{L}$. The difference is structural:

Table 6: PINN vs. homotopy: where physics enters.

Aspect	**PINN**	**Homotopy (this paper)**
Physics enters in	Loss function (penalty)	Operator $\mathcal{L}$ (structure)
Effect on architecture	None (still trial-and-error)	Reduces M constructively
Convergence	Gradient descent (linear)	Newton–Halley (quadratic/cubic)
Dimensioning	Not addressed	$\varepsilon \to M \to N$

Putting physics in the loss function (PINN) is adding a constraint. Putting physics in the operator (homotopy) is *reducing the problem*. The constraint does not change how many neurons are needed; the operator reduction does.

12.5 Implications for general AI architectures

Every neural network is a nonlinear operator $\mathcal{N} : \mathbb{R}^n \to \mathbb{R}^m$ parametrized by $\boldsymbol{\theta}$. The deformation equation (3) applies to all of them. The dimensioning chain (39) applies to all of them, with K_2 replaced by the appropriate measure of residual complexity after subtracting what $\mathcal{L}$ already knows.

Table 7 shows that all major neural architectures share the same bilinear structure: a last layer (or output head) that is *linear* in its weights, and internal parameters that are *nonlinear*. The homotopy framework exploits this universally: the linear weights are solved exactly by z_1 (one step), and the nonlinear parameters are corrected by z_1, z_2, z_3.

Table 7: Neural architectures as nonlinear operators with bilinear parameter structure. In all cases, the last-layer weights are linear (z_1 exact, one step) and the internal parameters are nonlinear (z_1, z_2, z_3 corrections).

Architecture	$\mathcal{N}[\phi]$	**Linear params**	**Nonlinear params**
RBF	$\sum w_j\, \varphi(\lVert x - c_j \rVert / \lambda_j)$	w_j	$c_j,\ \lambda_j$
MLP	$W_2\, \sigma(W_1 x + b_1)$	W_2 (last layer)	$W_1,\ b_1$
CNN	$W_{\text{fc}} \cdot \text{conv}(x;\, \mathcal{K})$	W_{fc}	kernels $\mathcal{K}$
LSTM	$W_o \cdot h_T(x;\, W_h)$	W_o	$W_h,\ W_f,\ W_i$
SOM	$\sum w_j\, \eta(\lVert x - m_j \rVert)$	topology (fixed)	prototypes m_j
Transformer	$W_{\text{head}} \cdot \text{attn}(x;\, Q, K, V)$	W_{head}	$Q,\ K,\ V$

For an MLP with activation σ, the Barron norm [24] $\lVert f \rVert_{\mathcal{B}}$ replaces K_2. The homotopy framework predicts:

$$M_{\text{MLP}} \propto \lVert \tilde{f} \rVert_{\mathcal{B}} = \lVert f - \mathcal{L}^{-1}[f] \rVert_{\mathcal{B}}. \tag{43}$$

Better prior knowledge (better $\mathcal{L}$) reduces $\lVert \tilde{f} \rVert_{\mathcal{B}}$ and therefore M.

This is the formal statement of what practitioners call "inductive bias": the choice of $\mathcal{L}$ *is* the inductive bias, and its quantitative effect on the architecture is given by the dimensioning chain. A CNN has implicit $\mathcal{L}$ = translation invariance. A Transformer has implicit $\mathcal{L}$ = attention to relevant tokens. The homotopy framework makes these implicit choices explicit and quantifiable.

13 Discussion

Why homotopy and not backpropagation? Backpropagation computes $\partial \mathcal{N} / \partial \boldsymbol{\theta}$ (the gradient) and takes a step of size η (learning rate) in the negative gradient direction. The homotopy correction $z_1 = -\mathcal{N}_0 / \mathcal{N}_0'$ is Newton's method—it uses the *same* Jacobian but divides by it instead of multiplying. Newton converges quadratically; gradient descent converges linearly. The Halley correction z_2 adds curvature information, reaching cubic convergence. No learning rate is needed because the step size is determined by the local geometry of $\mathcal{N}$.

Why separate linear and nonlinear parameters? The RBF residual is bilinear: linear in $\mathbf{w}$, nonlinear in $(\mathbf{c}, \boldsymbol{\lambda})$. Solving $\mathbf{w}$ exactly by linear algebra (one step) and applying homotopy only to $(\mathbf{c}, \boldsymbol{\lambda})$ reduces the nonlinear problem from $3M$ to $2M$ unknowns and exploits the bilinear structure optimally. This is the Variable Projection principle [8], but with Newton–Halley instead of gradient-based optimization on the nonlinear part.

Why the integral formulation for simulation? The BDF residual $g_{\text{BDF}} = (3y_k - 4y_{k-1} + y_{k-2})/(2T) + \hat{f}(y_k) - u_k$ contains $1/T$, amplifying sensor noise. The integral residual (20) contains

only T as a multiplicative factor, attenuating noise. Both have $O(T^2)$ discretization error, but the noise propagation properties differ fundamentally.

Limitations. The convergence of Level 1 (identification) depends on the K-means initialization being in the basin of attraction of the true centers. For highly multimodal problems, multiple restarts or global optimization may be needed for the initial centers. The convergence of Level 2 (simulation) requires T small enough that the 3-term homotopy series converges; for stiff systems, this may be restrictive.

Connection to the Homotopy Analysis Method. Liao's HAM [1] was developed for computing *analytic* series solutions to differential equations. The present work uses the HAM deformation equation as an *algorithmic* framework—the series is truncated at 3 terms and evaluated numerically at each step. The convergence-control parameter c_0 and the freedom of choosing $\mathcal{L}$ survive in the discrete setting as the parameter λ and the choice of kernel K.

14 Conclusions

1. Liao's deformation equation $(1-q)\mathcal{L}[\phi-\phi_0] = q\,c_0\,\mathcal{N}[\phi]$ is a universal root-finding framework. Its corrections z_1 (Newton), z_2 (Halley), z_3 (third order) are parametrized by (γ, σ) and apply to any $\mathcal{N}$.
2. For $\dot{y}+\mathrm{RBF}(y) = u(t)$, three problems—identification, simulation, and online refinement—are solved by the same corrections with different choices of $(\mathcal{L}, \mathcal{N}, \phi)$.
3. The bilinear structure of the RBF is exploited: weights are solved exactly (one linear step), centers and widths are corrected by Newton–Halley (3–5 iterations).
4. The linear starting solution $y_0(t) = y_0 e^{-kt} + \int_0^t e^{-k(t-s)} u(s)\,ds$ provides an analytically known initial approximation for the simulation homotopy, with kernel $K = e^{-k(t-s)}$ encoding the system's linear time constant.
5. The integral formulation for simulation eliminates numerical derivatives, with all residual coefficients at $O(1)$, ensuring noise robustness.
6. No gradients, no learning rates, no backpropagation. All corrections are closed-form.
7. The convergence conditions yield a constructive dimensioning chain: $\varepsilon_{\text{sim}} \to \varepsilon_{\text{id}} \to M_{\min} \to N_{\min} \to T_{\max}$. This answers the open question of how many neurons and how many data points are needed to guarantee a given simulation accuracy.

References

[1] S. J. Liao, *Beyond Perturbation: Introduction to the Homotopy Analysis Method*, Chapman & Hall/CRC, 2003.

[2] S. J. Liao, *Homotopy Analysis Method in Nonlinear Differential Equations*, Springer, 2012.

[3] S. J. Liao, "An optimal homotopy-analysis approach for strongly nonlinear differential equations," *Commun. Nonlinear Sci. Numer. Simul.*, vol. 15, pp. 2003–2016, 2010.

[4] S. Haykin, *Neural Networks and Learning Machines*, 3rd ed., Prentice Hall, 2009.

[5] J. Park, I. W. Sandberg, "Universal approximation using radial-basis-function networks," *Neural Comput.*, vol. 3, pp. 246–257, 1991.

[6] E. Hairer, S. P. Nørsett, G. Wanner, *Solving Ordinary Differential Equations I: Nonstiff Problems*, 2nd ed., Springer, 1993.

[7] L. Ljung, *System Identification: Theory for the User*, 2nd ed., Prentice Hall, 1999.

[8] G. H. Golub, V. Pereyra, "The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate," *SIAM J. Numer. Anal.*, vol. 10, pp. 413–432, 1973.

[9] Å. Björck, *Numerical Methods for Least Squares Problems*, SIAM, 1996.

[10] J. M. Gutiérrez, M. A. Hernández, "A family of Chebyshev–Halley type methods in Banach spaces," *Bull. Aust. Math. Soc.*, vol. 55, pp. 113–130, 1997.

[11] K. E. Atkinson, *The Numerical Solution of Integral Equations of the Second Kind*, Cambridge University Press, 1997.

[12] R. Kress, *Linear Integral Equations*, 3rd ed., Springer, 2014.

[13] R. H. Rodrigo, H. D. Patiño, S. J. Liao, "Integral homotopy regressors: unifying continuous and discrete HAM via compact Volterra operators," submitted to *Commun. Nonlinear Sci. Numer. Simul.*, 2026.

[14] R. H. Rodrigo, H. D. Patiño, "Homotopy-based functional neural network (HFNN) for nonlinear system identification," submitted to *Neurocomputing*, 2025.

[15] S. Chen, C. F. N. Cowan, P. M. Grant, "Orthogonal least squares learning algorithm for radial basis function networks," *IEEE Trans. Neural Netw.*, vol. 2, pp. 302–309, 1991.

[16] M. D. Buhmann, *Radial Basis Functions: Theory and Implementations*, Cambridge University Press, 2003.

[17] T. Poggio, F. Girosi, "Networks for approximation and learning," *Proc. IEEE*, vol. 78, pp. 1481–1497, 1990.

[18] D. S. Broomhead, D. Lowe, "Multivariable functional interpolation and adaptive networks," *Complex Syst.*, vol. 2, pp. 321–355, 1988.

[19] O. Nelles, *Nonlinear System Identification*, Springer, 2001.

[20] J. Sjöberg, Q. Zhang, L. Ljung, *et al.*, "Nonlinear black-box modeling in system identification: a unified overview," *Automatica*, vol. 31, pp. 1691–1724, 1995.

[21] K. S. Narendra, K. Parthasarathy, "Identification and control of dynamical systems using neural networks," *IEEE Trans. Neural Netw.*, vol. 1, pp. 4–27, 1990.

[22] G. Cybenko, "Approximation by superpositions of a sigmoidal function," *Math. Control Signals Syst.*, vol. 2, pp. 303–314, 1989.

[23] K. Hornik, "Approximation capabilities of multilayer feedforward networks," *Neural Netw.*, vol. 4, pp. 251–257, 1991.

[24] A. R. Barron, "Universal approximation bounds for superpositions of a sigmoidal function," *IEEE Trans. Inform. Theory*, vol. 39, pp. 930–945, 1993.

[25] M. Raissi, P. Perdikaris, G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Comput. Phys.*, vol. 378, pp. 686–707, 2019.